INDIAN INSTITUTE OF INFORMATION TECHNOLOGY RAICHUR
MATHEMATICS CLUB
PROBLEM NUMBER: WP2503
DATE: NOVEMBER 16, 2025
DUE DATE: NOVEMBER 30, 2025

WEEKLY PROBLEM

EXERCISE.

PART (A) — KEY COUNTING LEMMA

Let k be a positive integer. Consider sequences of length $2k+1$ made up of $k+1$ symbols '+1' and k symbols '-1'. A sequence is called **positive-prefix** if every nonempty prefix has a strictly positive sum (in the usual integer sense).

Prove that: the number of positive-prefix sequences equals

$$\frac{1}{2k+1}\binom{2k+1}{k}$$

Hint: *You may use the ballot theorem or the cycle lemma; a bijective or cyclic-rotation argument also works.*

A concrete example:

For $k=3$, the length of the sequence is $2(3)+1=7$. The sequence must contain four '+1's and three '-1's. The formula gives $\frac{1}{7}\binom{7}{3}=\frac{35}{7}=5$ valid sequences. 2 of the 5 valid sequences and their partial sums are $\longrightarrow$

$$++++---$$

Sums: 1, 2, 3, 4, 3, 2, 1

$$+++-+--$$

Sums: 1, 2, 3, 2, 3, 2, 1

PART (B) — THE MAIN PROBLEM

The integers:

$$1,2,3,\ldots,3k+1$$

are written down in a random order (each permutation being equally likely).

What is the probability that at no time during this process is the sum of the numbers written so far a positive integer divisible by 3?

(The answer is required in closed form)

Remark to the solver: Part (A) counts the admissible patterns of residues 1 and 2 (viewed as +1 and -1) after you remove the multiples of 3. Once you have that count, place the k zeros (multiples of 3) among the safe slots and multiply by the internal permutations of numbers with the same residue to finish Part (B).

(Scan this to submit solution!)

Top Submissions for last Week's Problem (WP2502):

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